PHYSICS

Higher Level

Set C | Paper 2

Challenge mock examination

Time allowed: 150 minutes Maximum mark: 90

Candidate name: _____

Candidate number: _____

Date: _____

Instructions

• A calculator and a clean current Physics data booklet are permitted throughout.

• All structured questions are compulsory. There are no optional topics or optional questions.

• Answer all questions. Show physical reasoning, working and units in written responses.

• Use the constants and assumptions given in a question. Otherwise use the data booklet and retain unrounded intermediate values.

• Give numerical answers to an appropriate number of significant figures. Diagrams are schematic unless explicitly stated to be to scale.

• The number in square brackets gives the maximum mark for a written part.

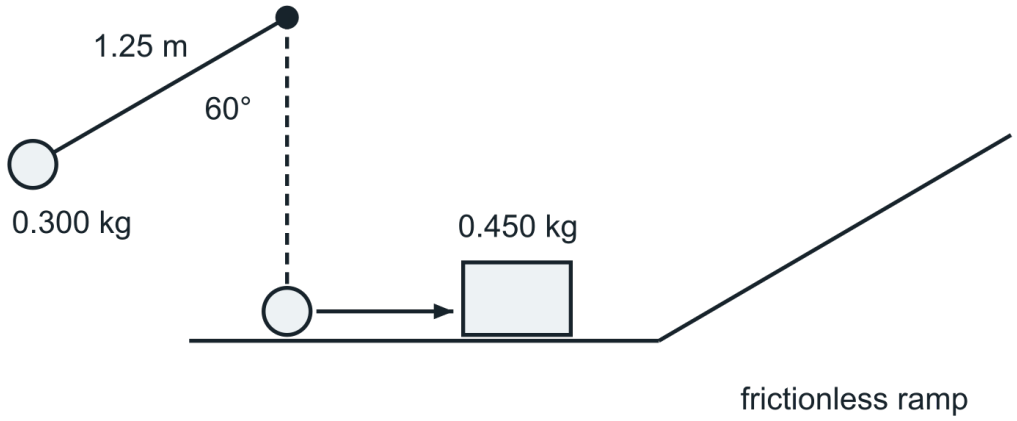
Thriving Scholars challenge practice | Current Physics syllabus | 2026

2 Question 1 | Question paper

Paper 2 - Structured questions

2 Question 1 [10 marks]

A $0.300\,\text{kg}$ pendulum bob on a light $1.25\,\text{m}$ string is released from rest at 60.0° to the downward vertical. At the lowest point the string is cut immediately before the bob collides horizontally with a stationary $0.450\,\text{kg}$ trolley. The bob sticks to the trolley, and the combined mass moves onto a frictionless ramp. Air resistance and rolling losses are negligible. Take $g = 9.81\,\text{m}\,\text{s}^{-2}$.



(a) Calculate the bob's speed immediately before the string is cut. [2]

(b) Calculate the string tension immediately before it is cut. [2]

(c) Determine the maximum vertical height reached by the combined bob and trolley, measured from the collision level. [3]

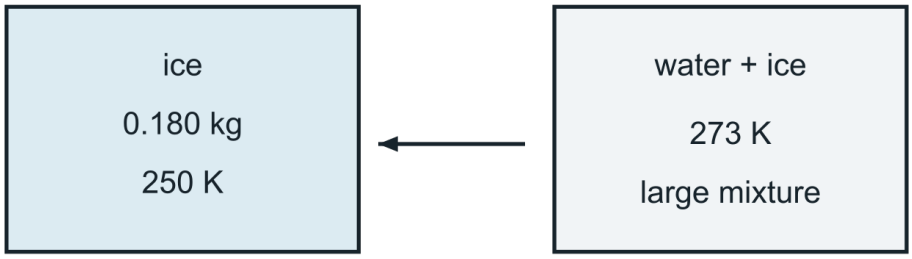
(d) Explain why horizontal momentum is conserved during the short collision although mechanical energy is not. [2]

2 Question 1 | Question paper

(e) Calculate the fraction of the bob's pre-collision kinetic energy dissipated in the collision. [1]

2 Question 2 [10 marks]

Ice of mass $0.180\,\text{kg}$ at $250\,\text{K}$ is put into a large stirred ice–water mixture at $273\,\text{K}$. The combined system is insulated, the container heat capacity is negligible, and enough liquid water remains that final equilibrium is at $273\,\text{K}$. Use ice specific heat capacity $c = 2100\,\text{J}\,\text{kg}^{-1}\,\text{K}^{-1}$ and latent heat of fusion $L_f = 3.34 \times 10^5\,\text{J}\,\text{kg}^{-1}$. For a solid with constant c, the entropy change between temperatures is supplied as $\Delta S = mc\ln(T_f/T_i)$.



Insulated combined system; negligible container heat capacity

(a) Determine the mass of liquid water that freezes. [2]

(b) Explain how the freezing water loses internal energy while its temperature remains constant. [2]

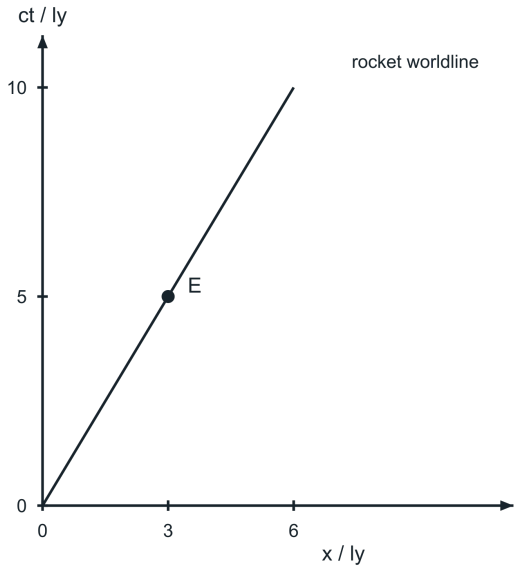
(c) Calculate the entropy increase of the initially cold ice. [2]

(d) Calculate the total entropy change of the combined system and explain its consistency with the second law. [2]

(e) In a repeat, the same ice mass starts at $270\,\text{K}$. Calculate the new total entropy change and explain why it differs from part (d). [2]

2 Question 3 [10 marks]

A rocket passes a space station at event O and then moves in the positive x direction at constant speed. The diagram uses the station frame S, with x and ct both in light-years. Event E on the rocket has coordinates $(x, ct) = (3.0, 5.0)\,\text{ly}$. At E the rocket emits a photon towards the station, which remains at $x = 0$. The rocket frame S′ shares event O as its origin. Use $x' = \gamma(x - \beta ct)$ and $ct' = \gamma(ct - \beta x)$, where $\beta = v/c$.



(a) Determine β and γ for the rocket. [2]

(b) Determine the station-frame coordinates of the photon's arrival F and the spacetime interval $s^2 = (\Delta ct)^2 - (\Delta x)^2$ between E and F. [2]

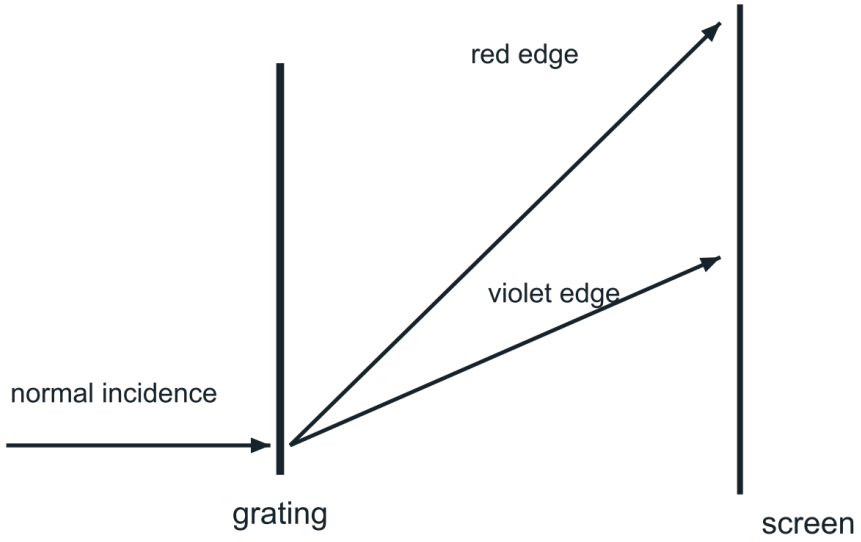
(c) Transform E into S′ and identify the elapsed time on the rocket clock from O to E. [2]

(d) Transform F into S′. Explain why its time coordinate is not a reading made by the rocket's clock at F. [2]

(e) Event G occurs at the station and is simultaneous with E in S. Determine whether G is earlier or later than E in S′, and by how much. [2]

2 Question 4 [10 marks]

A diffraction grating has $500\,\text{lines}\,\text{mm}^{-1}$. Light covering all wavelengths from $400\,\text{nm}$ to $700\,\text{nm}$ is incident normally. A distant flat screen is perpendicular to the undeviated beam. Use $d\sin\theta = m\lambda$.



(a) Calculate the grating spacing in metres. [1]

(b) Calculate the angular width of the complete second-order spectrum. [2]

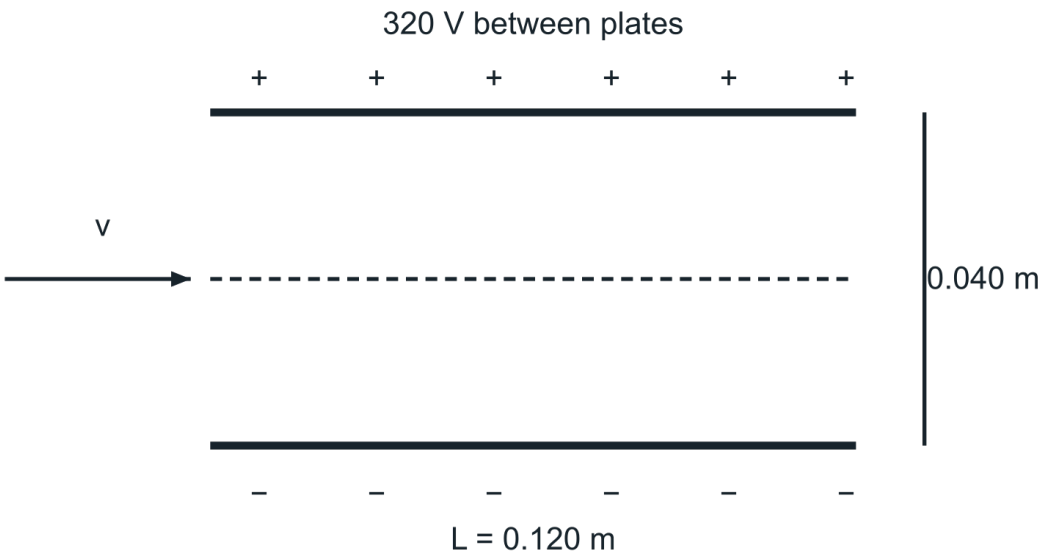
(c) Determine the highest order containing the whole incident wavelength range, explaining the limiting wavelength. [2]

(d) At one screen position, second-order $660\,\text{nm}$ light overlaps third-order light. Determine the third-order wavelength and the diffraction angle. [3]

(e) The source is replaced by monochromatic $520\,\text{nm}$ light. Determine the total number of diffraction maxima, including both sides and the central maximum, that can reach the screen. [2]

2 Question 5 [10 marks]

Horizontal plates of length $0.120\,\text{m}$ are separated by $0.0400\,\text{m}$ in a vacuum. Their potential difference is $320\,\text{V}$, with the upper plate positive. A proton enters midway between them with horizontal velocity $3.00 \times 10^{5}\,\text{m}\,\text{s}^{-1}$. Edge fields and gravity are negligible. Use $e = 1.60 \times 10^{-19}\,\text{C}$ and $m_p = 1.67 \times 10^{-27}\,\text{kg}$.



(a) Calculate the electric field strength and the proton's vertical acceleration, including directions. [2]

(b) Determine whether the proton reaches the far end of the plates. If it hits a plate first, determine the horizontal distance to impact. [3]

(c) An alpha particle, of charge $+2e$ and mass $4m_p$, enters at the same point and speed. Determine whether it hits a plate. [2]

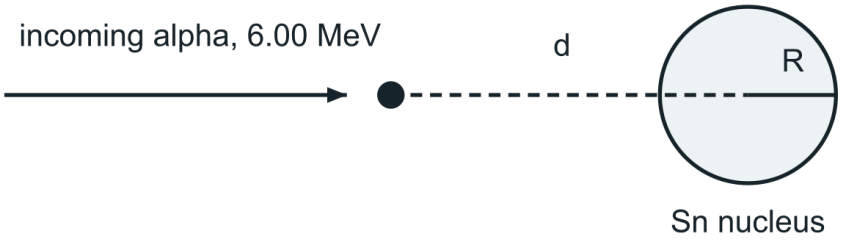
(d) A uniform magnetic field is added so that either ion at the stated speed passes undeflected. Determine its magnitude and direction. [2]

2 Question 5 | Question paper

(e) For an undeflected ion in the combined fields, state the net change in kinetic energy while it passes between the plates, with a reason. [1]

2 Question 6 [10 marks]

An alpha particle approaches a stationary tin nucleus head-on with initial kinetic energy $6.00\,\text{MeV}$ when far away. The tin proton number is 50, and its nucleon number is 120. Neglect target recoil and assume electrostatic repulsion outside the nucleus. Use $ke^2 = 1.44\,\text{MeV fm}$ and $R = 1.20A^{1/3}\,\text{fm}$.



Head-on; heavy target recoil neglected

(a) Determine the distance of closest approach between the nuclear centres. [2]

(b) Calculate the tin nuclear radius and the ratio of closest approach to this radius. [2]

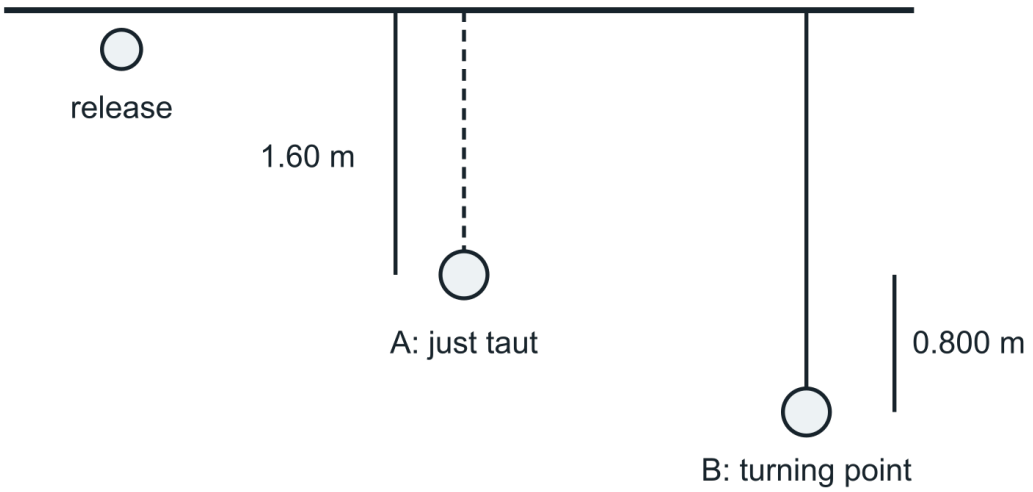
(c) The target is replaced by another tin isotope of nucleon number 125. Explain how the predicted closest approach and nuclear radius change at the same incident alpha energy. [2]

(d) Estimate the alpha energy for which the electrostatic closest approach would equal the radius of the original $A = 120$ nucleus. [2]

(e) Explain two reasons why the point-charge electrostatic description may fail once the closest-approach distance becomes comparable with the nuclear radius. [2]

2 Question 7 [10 marks]

A $2.00\,\text{kg}$ test mass is released from rest at the fixed end of a light elastic safety cord. The cord is slack until the mass has fallen its unstretched length $1.60\,\text{m}$. It then obeys Hooke's law and reaches maximum extension $0.800\,\text{m}$. Energy losses are negligible; use $g = 10.0\,\text{m}\,\text{s}^{-2}$. A timer measures $0.205\,\text{s}$ from the cord first becoming taut at A to the mass stopping at B.



(a) Calculate the speed at A. [2]

(b) Determine the cord's spring constant. [2]

(c) Determine the magnitude and direction of the acceleration at B. Explain why zero speed does not imply equilibrium. [2]

(d) Use impulse to calculate the average upward cord force between A and B. [2]

(e) A student estimates this average force as one half of the maximum cord force. Test that estimate and explain why a linear force–extension law does not justify it. [2]

2 Question 8 [10 marks]

Two identical vertically oscillating drivers generate coherent water waves at S_1 and S_2. Each driver is a mass $0.160\,\text{kg}$ on a spring of constant $64\pi^2\,\text{N m}^{-1}$. Initially they oscillate in phase at their natural frequency. The wave speed is constant at $1.50\,\text{m s}^{-1}$. At P the source distances are $2.40\,\text{m}$ and $1.95\,\text{m}$. Each wave alone has displacement amplitude $0.800\,\text{mm}$ at the points considered; ignore damping and attenuation.

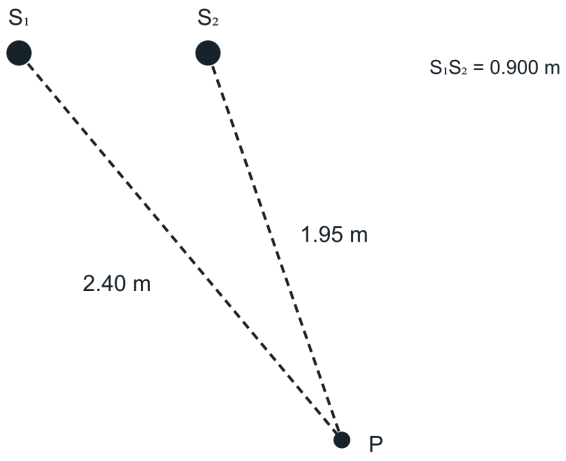


Diagram not to scale

(a) Calculate the initial frequency of the drivers. [2]

(b) Calculate the water wavelength. [1]

(c) Determine the resultant amplitude at P, explaining the interference condition. [2]

(d) The drivers remain in phase while their common frequency is increased continuously. Determine the first frequency above the initial value for which P is a displacement node. [3]

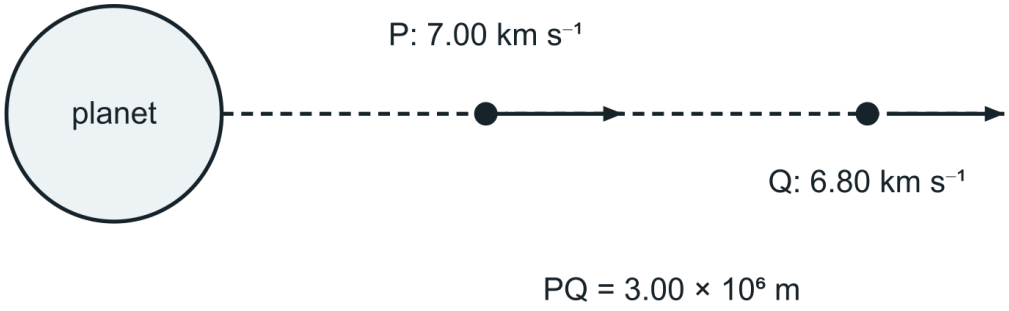
2 Question 8 | Question paper

(e) The initial frequency is restored, then one driver's phase is reversed by half a cycle. State the resultant amplitudes at P and at Q, where the path difference is $0.525\,\text{m}$, explaining the conditions. [2]

2 Question 9 [10 marks]

A non-relativistic probe leaves an airless planet with speed $10.0\,\mathrm{km\,s^{-1}}$. The escape speed from the launch point is $8.00\,\mathrm{km\,s^{-1}}$. After launch only the planet's gravity acts. Farther out on the same radial path, its speed decreases from $7.00\,\mathrm{km\,s^{-1}}$ at P to $6.80\,\mathrm{km\,s^{-1}}$ at Q over $3.00\times10^{6}\,\mathrm{m}$.

A separate helium-burning stellar core supports the reaction $3\,{}^{4}_{2}\mathrm{He}\rightarrow{}^{12}_{6}\mathrm{C}+\gamma$. Use atomic masses $m({}^{4}\mathrm{He})=4.00260\,\mathrm{u}$, $m({}^{12}\mathrm{C})=12.00000\,\mathrm{u}$ and $1\,\mathrm{u}\,c^{2}=931.5\,\mathrm{MeV}$. The atomic electron counts balance between reactants and products.



(a) Determine the probe's speed at a very large distance from the planet. [2]

(b) Determine the magnitude of the gravitational field averaged along the path from P to Q, using work and energy. [2]

(c) Calculate the energy released in the stated fusion reaction. [2]

(d) Explain why helium fusion requires higher core temperatures than fusion of hydrogen nuclei, and why fusion beyond the iron-group peak cannot provide a sustained stellar energy source. [2]

2 Question 9 | Question paper

(e) Outline the mechanical condition for the stellar core to remain in hydrostatic equilibrium and how nuclear energy generation helps maintain that condition. [2]

End of question paper

PHYSICS

Higher Level

Set C | Paper 2

Worked solution book

Complete the examination before consulting this book. Marking points indicate a method (M), accuracy (A) or reasoning (R) award. The number after the code gives its value. Award only points supported by the candidate's retained working.

Accept valid equivalent approaches and reasonable rounding. Apply error carried forward only where the subsequent method remains physically meaningful; do not reward a contradiction of stated conditions. A correct multiple-choice letter earns one mark; no negative marks apply.

MCQs are automatically marked on the portal. Written marks are self-assessed. Reports show these separately and do not calculate an official IB grade.

Paper 2 - Worked solutions

2 Question 1 [10 marks]

(a) [2 marks]

M1 The vertical drop is $h = L(1 - \cos 60^\circ) = 0.625\,\text{m}$, and $mgh = \tfrac{1}{2}mv^2$.

A1 $v = \sqrt{2(9.81)(0.625)} = 3.50\,\text{m s}^{-1}$.

(b) [2 marks]

M1 At the bottom, $T - mg = mv^2/L$; centripetal force is the resultant, not an additional force.

A1 $T = 0.300[9.81 + (3.5018)^2/1.25] = 5.89\,\text{N}$.

(c) [3 marks]

M1 Horizontal momentum gives $0.300v = (0.300 + 0.450)u$.

A1 $u = 1.40\,\text{m s}^{-1}$ after sticking.

A1 On the lossless ramp, $\tfrac{1}{2}Mu^2 = MgH$, so $H = u^2/(2g) = 0.100\,\text{m}$. Accept a consistent follow-through from part (a).

(d) [2 marks]

R1 The string has been cut and external horizontal impulse during the short collision is negligible; the bob–trolley system's horizontal momentum is conserved.

R1 Sticking is inelastic: mechanical energy transfers to internal energy/deformation/sound. These transfers do not require loss of total energy.

(e) [1 mark]

A1 $K_f/K_i = m_{\text{bob}}/(m_{\text{bob}} + m_{\text{trolley}}) = 0.400$, so the dissipated fraction is 0.600.

2 Question 2 [10 marks]

(a) [2 marks]

M1 The cold ice requires $Q = mc(273 - 250) = 0.180(2100)(23) = 8694\,\text{J}$. The freezing water supplies this latent energy.

A1 $m_f = Q/L_f = 8694/(3.34 \times 10^5) = 0.0260\,\text{kg}$.

(b) [2 marks]

R1 Temperature reflects mean random kinetic energy, which does not decrease during the equilibrium phase change.

R1 Intermolecular potential energy decreases as the solid structure forms; the released latent energy warms the initially colder ice. Thus constant temperature does not imply constant internal energy.

(c) [2 marks]

M1 Apply the supplied relationship to the warming ice, not $Q/273$ for the whole varying-temperature process.

A1 $\Delta S_{\text{ice}} = 0.180(2100)\ln(273/250) = 33.3\,\text{J K}^{-1}$.

(d) [2 marks]

A1 The water freezes at $273\,\text{K}$, so $\Delta S_{\text{water}} = -8694/273 = -31.8\,\text{J K}^{-1}$. Total entropy increases by about $1.42\,\text{J K}^{-1}$.

R1 The isolated combined system has positive entropy production from energy transfer across a finite temperature difference; one component may lose entropy while the total increases.

(e) [2 marks]

A1 $\Delta S_{\text{total}} = 0.180(2100)[\ln(273/270) - 3/273] \approx 0.0230\,\text{J K}^{-1}$.

R1 The initial temperature difference is much smaller, so the energy transfer is closer to reversible and produces less entropy. Both totals remain positive.

2 Question 3 [10 marks]

(a) [2 marks]

A1 $\beta = x/(ct) = 3.0/5.0 = 0.600.$

A1 $\gamma = (1 - 0.600^2)^{-1/2} = 1.25.$

(b) [2 marks]

A1 The backward photon travels $3.0\,\text{ly}$, so $F : (x, ct) = (0, 8.0)\,\text{ly}$.

A1 $\Delta ct = 3.0\,\text{ly}$ and $\Delta x = -3.0\,\text{ly}$, so $s^2 = 0\,\text{ly}^2$; the interval is lightlike.

(c) [2 marks]

A1 $x'_E = 1.25(3 - 0.6 \times 5) = 0$, and $ct'_E = 1.25(5 - 0.6 \times 3) = 4.0\,\text{ly}$.

A1 The rocket clock reads an elapsed proper time $4.0\,\text{yr}$ because O and E both occur at $x' = 0$.

(d) [2 marks]

A1 $x'_F = 1.25(0 - 0.6 \times 8) = -6.0\,\text{ly}$ and $ct'_F = 1.25(8 - 0) = 10.0\,\text{ly}$.

R1 F is at $x' = -6.0\,\text{ly}$, not at the rocket clock. Its time is the S′ coordinate time assigned by synchronized clocks in that frame; the rocket clock does not pass through F.

(e) [2 marks]

M1 $G : (x, ct) = (0, 5.0)\,\text{ly}$, so $ct'_G = 1.25(5) = 6.25\,\text{ly}$.

A1 G is later than E by $(6.25 - 4.0)\,\text{yr} = 2.25\,\text{yr}$ in S′, despite simultaneity in S.

2 Question 4 [10 marks]

(a) [1 mark]

A1 $d = 1/(500 \times 10^3) = 2.00 \times 10^{-6}\,\text{m}$.

(b) [2 marks]

M1 $\theta_{\text{violet}} = \sin^{-1}[2(400 \times 10^{-9})/d] = 23.58^\circ$ and $\theta_{\text{red}} = \sin^{-1}[2(700 \times 10^{-9})/d] = 44.43^\circ$.

A1 The angular width is $44.43^\circ - 23.58^\circ = 20.85^\circ$. The width is a difference of inverse-sine angles, not the inverse sine of a wavelength difference.

(c) [2 marks]

M1 The longest wavelength is most restrictive: $m \le d/(700 \times 10^{-9}) = 2.857$.

A1 The highest complete order is $m = 2$. Some shorter wavelengths appear in order 3, but the entire range does not.

(d) [3 marks]

M1 At the same angle, $2\lambda_2 = 3\lambda_3$.

A1 $\lambda_3 = 2(660)/3 = 440\,\text{nm}$.

A1 $\theta = \sin^{-1}[2(660 \times 10^{-9})/d] = 41.30^\circ$.

(e) [2 marks]

M1 $m_{\max} = \lfloor d/(520 \times 10^{-9}) \rfloor = 3$; no grazing order is at the equality boundary for these data.

A1 The total is $2m_{\max} + 1 = 7$ maxima, at orders $0, \pm1, \pm2, \pm3$.

2 Question 5 [10 marks]

(a) [2 marks]

A1 $E = V/s = 320/0.0400 = 8.00 \times 10^{3}\ \text{V m}^{-1}$, downwards from the positive plate.

A1 $a_p = eE/m_p = 7.66 \times 10^{11}\ \text{m s}^{-2}$, downwards.

(b) [3 marks]

M1 The nominal transit time is $L/v = 4.00 \times 10^{-7}\ \text{s}$; unbounded vertical deflection would be $\tfrac{1}{2}a_p t^2 = 0.0613\ \text{m} > 0.0200\ \text{m}$, so the lower plate is hit.

M1 Set $0.0200 = \tfrac{1}{2}a_p t_h^2$ to obtain $t_h = 2.28 \times 10^{-7}\ \text{s}$.

A1 $x_h = vt_h = 0.0685\ \text{m}$, less than $0.120\ \text{m}$. The nominal-exit deflection is a test, not an actual path beyond impact.

(c) [2 marks]

M1 $a_\alpha = (2e)/(4m_p)\,E = a_p/2$. Its nominal-exit deflection is $0.0307\ \text{m}$.

A1 This still exceeds $0.0200\ \text{m}$, so the alpha also hits the lower plate, at $x = \sqrt{2}\,x_h \approx 0.0969\ \text{m}$.

(d) [2 marks]

A1 $qE = qvB$ gives $B = E/v = 0.0267\ \text{T}$, independent of ion charge-to-mass ratio.

R1 B must be into the page: for positive charge moving right, the magnetic force is then upward and opposes the downward electric force.

(e) [1 mark]

R1 The kinetic-energy change is zero. The net force vanishes, and the electric force is perpendicular to the straight trajectory; the magnetic force does no work.

2 Question 6 [10 marks]

(a) [2 marks]

M1 At the head-on turning point, relative kinetic energy is zero and $K = k(2e)(50e)/d$.

A1 $d = 2(50)(1.44)/6.00 = 24.0\,\text{fm}$. This is centre-to-centre distance.

(b) [2 marks]

A1 $R = 1.20(120)^{1/3} = 5.92\,\text{fm}$.

A1 $d/R = 24.0/5.92 = 4.05$.

(c) [2 marks]

R1 Closest approach is unchanged in the stated no-recoil model because both isotopes have Z=50 and the incident energy is unchanged.

R1 Radius increases by $(125/120)^{1/3} = 1.014$, about 1.4%, because nuclear radius depends on nucleon number.

(d) [2 marks]

M1 Set $d = R$ in the electrostatic energy relation, giving $K_{\text{est}} = 2Zke^2/R$.

A1 $K_{\text{est}} = 144/5.92 = 24.3\,\text{MeV}$. It is an estimate of a scale, not an exact threshold for every nuclear effect.

(e) [2 marks]

R1 The target charge occupies a finite volume, so when the alpha penetrates that volume the outside point-charge potential no longer applies.

R1 At nuclear separations the short-range strong nuclear interaction becomes important and electrostatic repulsion alone is insufficient. Accept a specific valid discussion of nuclear response/reactions; do not award a second mark for merely repeating that nuclei have a size.

2 Question 7 [10 marks]

(a) [2 marks]

M1 Before A the mass falls freely, so $v_A^2 = 2gL$.

A1 $v_A = \sqrt{2(10.0)(1.60)} = 5.66\,\text{m s}^{-1}$ downwards.

(b) [2 marks]

M1 The full fall to B is $L + x = 2.40\,\text{m}$, so $mg(L + x) = \tfrac{1}{2}kx^2$.

A1 $k = 2(2.00)(10.0)(2.40)/(0.800)^2 = 150\,\text{N m}^{-1}$.

(c) [2 marks]

A1 At B, $F_{\text{cord}} = kx = 120\,\text{N}$ upward, giving resultant force $120 - 20 = 100\,\text{N}$ and acceleration $50.0\,\text{m s}^{-2}$ upward.

R1 B is an instantaneous turning point. Its velocity is zero but the forces do not balance, so it is not an equilibrium state.

(d) [2 marks]

M1 Take upward positive: $(\overline{F}_{\text{cord}} - mg)\Delta t = m[0 - (-v_A)]$.

A1 $\overline{F}_{\text{cord}} = 20.0 + 2.00(5.6569)/0.205 = 75.2\,\text{N}$. Accept a consistent follow-through from part (a).

(e) [2 marks]

A1 Half the maximum force is $60.0\,\text{N}$, which does not match the impulse-derived $75.2\,\text{N}$.

R1 Hooke’s law makes force linear in extension, not linear in time. The changing velocity means extensions do not occupy equal time intervals, so the time average is not the midpoint of the force range.

2 Question 8 [10 marks]

(a) [2 marks]

M1 $f = (1/2\pi)\sqrt{k/m}$ for each ideal mass–spring driver.

A1 $f = (1/2\pi)\sqrt{64\pi^2/0.160} = 10.0\,\text{Hz}$.

(b) [1 mark]

A1 $\lambda = v/f = 1.50/10.0 = 0.150\,\text{m}$.

(c) [2 marks]

R1 Path difference is $2.40 - 1.95 = 0.450\,\text{m} = 3\lambda$, so initially in-phase waves arrive in phase.

A1 Amplitudes add, giving $A_P = 1.60\,\text{mm}$.

(d) [3 marks]

M1 For in-phase sources a node requires $\Delta r = (n + \tfrac{1}{2})\lambda = (n + \tfrac{1}{2})v/f$.

R1 The initial path difference is three wavelengths. Increasing frequency increases $\Delta r/\lambda$, so the next node is at 3.5, not 2.5 wavelengths.

A1 $f = 3.5(1.50)/0.450 = 11.7\,\text{Hz}$.

(e) [2 marks]

R1 At P, the propagation phase difference is 6π; adding the source phase reversal gives destructive interference, $A_P = 0$.

R1 At Q, $\Delta r/\lambda = 0.525/0.150 = 3.5$. Its odd-π propagation phase difference is cancelled by the source phase reversal, so $A_Q = 1.60\,\text{mm}$.

2 Question 9 [10 marks]

(a) [2 marks]

M1 With potential zero at infinity, the energy required for escape is $\frac{1}{2}mv_{\text{esc}}^2$. Thus $\frac{1}{2}mv_0^2 - \frac{1}{2}mv_{\text{esc}}^2 = \frac{1}{2}mv_\infty^2$.

A1 $v_\infty = \sqrt{10.0^2 - 8.00^2} = 6.00\,\text{km s}^{-1}$.

(b) [2 marks]

M1 The positive loss of kinetic energy per unit mass equals $\overline{g}\,s$: $\overline{g} = (v_P^2 - v_Q^2)/(2s)$. This defines the path average even though g is not constant.

A1 $\overline{g} = [7000^2 - 6800^2]/[2(3.00 \times 10^6)] = 0.460\,\text{N kg}^{-1}$ towards the planet.

(c) [2 marks]

M1 The mass defect is $3(4.00260) - 12.00000 = 0.00780\,\text{u}$; balanced electron counts permit use of the supplied atomic masses.

A1 $E = 0.00780(931.5) = 7.27\,\text{MeV}$ per reaction.

(d) [2 marks]

R1 Helium nuclei have greater positive charge than hydrogen nuclei, hence greater electrostatic repulsion; higher random kinetic energies/temperatures are required to approach closely enough for nuclear fusion.

R1 Near the iron-group binding-energy-per-nucleon maximum, further fusion no longer gives a net increase in binding energy per nucleon and becomes energy-consuming, so it cannot sustain energy output.

(e) [2 marks]

R1 The outward pressure-gradient force balances inward gravity locally; accept thermal and/or radiation pressure as appropriate.

R1 Nuclear energy replenishes energy radiated away, maintaining core temperature and pressure support. Equilibrium does not mean gravity vanishes or that no energy flows outward.

End of worked solution book